Nguyen Khoi Nguyen (Alan)

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Experience

Junior Researcher · *DCL & H2X at Boston University* — *Sep 2025 – Present · Boston, MA (On-site)*

- Full-time research on robot learning and machine perception with Prof. Wenchao Li (Dependable Computing Lab) and Prof. Eshed Ohn-Bar (H2X Lab): end-to-end autonomous driving in CARLA and improving the generalization and reliability of vision-language models (VLMs) and vision-language-action models (VLAs) across simulated and real-world robotic platforms.
- Built the 3D perception stack for a 7-DoF manipulator (camera calibration, point clouds, segmentation, grasp pose estimation, ROS1/ROS2, Gazebo/Isaac Sim); ran a 400-trial benchmark of vision-language models as task planners and reinforcement-learning benchmarking in simulation (Stable-Baselines3).

Principal Software Engineer · *Crown Barber Studio* — *Oct 2025 – Apr 2026 · Palo Alto, CA (Remote Contract)*

- Sole engineer on a startup 3D hair-scanning product: built the entire iOS app (ARKit capture, neural 3D reconstruction, on-device rendering) and its cloud API (FastAPI, MongoDB, Firebase); shipped to the App Store, credited as developer.

Summer Intern, Physical Data & Post-Training · *Scale* — *Summer 2025 · San Francisco, CA (Hybrid)*

- Developed AI training and evaluation procedures for computational physics (finite element, FDTD, beam propagation); built benchmarking suites in Python, C++, and MATLAB.

Undergraduate Researcher · *CISL at Boston University* — *Feb 2025 – Sep 2025 · Boston, MA (On-site)*

- Researched intensity diffraction tomography with Prof. Lei Tian at the Computational Imaging Systems Lab (3D refractive-index reconstruction from microscopy): transfer-function derivations, Python forward models, and Fourier-domain fidelity analysis.

Summer Intern, Data Science and Computer Vision · *Leuko* — *Summer 2023 · Boston, MA (On-site)*

- Analyzed blood samples and developed OpenCV-based computer vision pipelines to detect and characterize blood cells from optical microscopy; built dataset versioning tooling for labeled biomedical image sets.

Technical Skills

ML, RL & CUDA	PyTorch, scikit-learn, Hugging Face, Stable-Baselines3, Gymnasium, CUDA; LLM/VLM-based pipelines and agents, evaluation design, statistical model selection
Perception & 3D	OpenCV, Open3D, SAM, GroundingDINO, ARKit, OpenGL, Blender; camera calibration, 3D reconstruction, pose estimation, segmentation, microscopy imaging
Robotics & Simulation	ROS1/ROS2, CARLA, MuJoCo, Gazebo, Isaac Sim, LIBERO
Programming	Python (NumPy, SciPy), C/C++, Rust, Julia, SQL, JavaScript, MATLAB, VHDL, Assembly, YAML, LaTeX
Cloud & MLOps	AWS, Docker, Linux, Git, CI/CD, FastAPI, React, PostgreSQL, MongoDB, Firebase; model validation, production monitoring

Education

B.A. Physics & Computer Science (Dual Major) · *Boston University* — *Sep 2020 – Jul 2025 · Boston, MA*

Relevant Coursework: Machine Learning (CS542), Deep Learning (CS523), Image & Video Computing (CS585), Software Engineering (CS411), Computational Optical Imaging (EC522), Cyber-Physical Systems (EC545); plus graduate coursework in photonics and optics (EC 562/565/591/763/777).

Leadership & Distinctions

Co-Founder & President, **Boston University AI Society** — *Jan 2023 – Sep 2025*

Led BU's largest student-run AI community; raised $10,000+ for AI and robotics hardware; advised BU Student Government on AI policy (TerrierGPT). Graduate TA for EC 574/575; lead TA, EC 545 Cyber-Physical Systems.

Vietnamese national team reserve, 2018 International Mathematical Olympiad (did not attend)